Semester	**IV**
Subject Code	**23AD411**
Subject Title	**Machine Learning Techniques**
Duration	One Semester
Total Contact Hours	45 Hours per semester
Pre-Requisite	23FYM21 - Linear Algebra and Gradient Calculus 23AD31 – Probability and Applied Statistics
Credit Points	3 CP
Learning and Teaching Structure	<table><tr><td>L</td><td>T</td><td>P</td><td>C</td></tr><tr><td>3</td><td>0</td><td>0</td><td>3</td></tr></table>
Assessment Type	Theory
Course Objective	To provide an understanding of the theoretical concepts of machine learning and preparestudents for research or industry application of machine learning techniques.

Course Outcome

CO1: To understand the basic principles of machine learning techniques

CO2: To evaluate the strengths and weaknesses of machine learning algorithms

CO3: To Study the concepts of dimensionality reduction

CO4: To Summarize the concepts in Bayesian analysis from probability models and methods

CO5: To Design and implement neural network and clustering algorithms

CO6: To Understand the basics of computation and ensemble learning techniques

INTRODUCTION TO MACHINE LEARNING

Introduction- Types of Machine Learning- Application of Machine Learning- Hypothesis Space -Inductive Bias- Evaluation and Cross Validation. **(9)**

SUPERVISED LEARNING ALGORITHMS

Linear Models for Regression – Linear Models for Classification- Discriminant Functions, Probabilistic Generative Models, Probabilistic Discriminative Models – Decision Tree Learning – Bayesian Learning, Naïve Bayes –Ensemble Methods, Bagging, Boosting. **(9)**

UNSUPERVISED LEARNING ALGORITHMS

Clustering- Hierarchial Agglomerative Clustering-K-means – EM Algorithm- Mixtures of Gaussians –Dimensionality Reduction, Linear Discriminant Analysis, Factor Analysis, Principal Components Analysis, Independent Components Analysis. **(9)**

NEURAL NETWORK AND BAYESIAN CONCEPTS
Introduction to Neural Network- Biological Neurons- Architectural of Neural Network-Implementation of ANN -Back propagation Algorithm – Importance of Bayesian Methods: Bayes Theorem - Bayes Theorem and Concept -Learning- Bayes Belief Network **(9)**
CLUSTERING AND COMPUTATION LEARNING
Introduction to Clustering,Hierarchical Clustering -Agglomerative Clustering. Introduction to Computation Learning -Sample complexity: Finite Hypothesis Space- VC Dimensions- Introduction to Ensemble Learning **(9)**
TOTAL: 45
Minimum Requirement to pass the subject
Text Book 1. I. A. Dhotre , Introduction to Machine learning, Technical Publication, 2022 2. T. Mitchell, Machine Learning, McGraw Hill, 1997.
Reference Book 1. I. Goodfellow, Y. Bengio and A. Courville. Deep Learning. MITPress, 2016. 2. Richard S. Sutton and Andrew G. Barto, Reinforcement Learning:An Introduction, MIT Press, 2018. 3. M. P. Deisenroth, A. A. Faisal, C. S. Ong, Mathematics forMachine Learning, Cambridge University Press (1st edition), 2020. 4. S. Haykin. Neural networks and learning machines. Pearson 2008. 5. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 6. K. Murphy. Machine Learning: A probabilistic perspective, MIT Press, 2012. 7. Hastie, Tibshirani, Friedman, Elements of statistical learning, Springer, 2011.